

DAILY IT TRAINING

EVE-NG Airport VLAN Network Project

Interview Explanation Guide for HR and IT Hiring Managers

Area	Networking / NOC / Network Support Portfolio
Subtopic	VLANs, trunking, router-on-a-stick, DHCP, DNS, ACLs, SSH, troubleshooting, and
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What's Inside

- How I Would Introduce This Project
- Business Scenario: Why This Project Matters
- Technology Overview
- Step-by-Step Project Walkthrough
- Step 1: I identified the departments and staff types
- Step 2: I created a VLAN and IP addressing plan
- Step 3: I configured VLANs on the core switch
- Step 4: I assigned switch ports as access ports
- Step 5: I configured the switch-to-router trunk
- Step 6: I configured router-on-a-stick
- Step 7: I created DHCP pools per VLAN
- Step 8: I added an internal DNS design
- Step 9: I configured ACLs for security
- Step 10: I hardened management access with SSH
- Appendix: Each Networking Concept

1. How I Would Introduce This Project in an Interview

If HR or an IT hiring manager asked me, "Tell me about your VLAN project," I would explain it like this:

> I built an airport-style network lab in EVE-NG to simulate how different airport departments connect securely on one shared network infrastructure. The lab includes separate VLANs for check-in, gate operations, baggage devices, airport operations, IT management, guest Wi-Fi, and internal servers. I used a Layer 2 switch for VLAN access ports, an 802.1Q trunk between the switch and router, router-on-a-stick for inter-VLAN routing, DHCP pools for automatic IP addressing, internal DNS for service-name resolution, ACLs for security control, and SSH hardening for secure device management. The most valuable part of the project was troubleshooting a real DHCP problem caused by ACLs blocking DHCP Discover traffic, then fixing it and verifying the result.

In simple terms, this project shows that I can design, configure, secure, verify, and troubleshoot a small enterprise network using real Cisco-style networking concepts.

2. Business Scenario: Why This Project Matters

The scenario is a regional airline or airport operations network. In an airport, different groups use the network for very different purposes:

- Check-in staff need stable access to airline systems.
- Gate staff need boarding and passenger-service access.
- Baggage scanners and baggage printers support flight operations.
- Airport operations staff need internal application access.
- IT administrators need secure management access to routers and switches.
- Guest Wi-Fi users need internet-style access but should not touch internal systems.
- Servers provide internal services such as DNS and airport applications.

The business problem is that all these users may physically connect through the same network equipment, but they should not all be treated the same. Guest traffic should not mix with baggage systems. Baggage devices should not have unlimited access. IT management should be protected. That is why I designed the network with segmentation and security controls.

3. Topology Overview

The lab uses a simple but realistic enterprise topology:

```

R1-AIRPORT-EDGE
Router-on-a-stick / DHCP / ACLs / SSH management
FastEthernet0/0
|
| 802.1Q trunk carrying VLANs 10,20,30,40,50,60,70
|
SW1-CORE
Layer 2 access switch
|
+-- VLAN 10 CHECKIN          - check-in PCs
+-- VLAN 20 GATE             - gate operations PCs
+-- VLAN 30 BAGGAGE          - scanners and printers
+-- VLAN 40 AIRPORT_OPS      - operations PCs
+-- VLAN 50 IT_MGMT          - admin/IT workstation
+-- VLAN 60 GUEST_WIFI       - guest users
+-- VLAN 70 SERVERS          - DNS and application servers

```

The switch handles Layer 2 connectivity and VLAN membership. The router handles Layer 3 gateways, routing between VLANs, DHCP, ACL security, and management access.

4. Step-by-Step Project Walkthrough

Step 1: I identified the departments and traffic types

Before configuring anything, I separated the airport environment into business groups. This helped me decide which VLANs and subnets were needed.

Step 2: I created a VLAN and IP addressing plan

Each department received its own VLAN and /24 subnet. This made the design clean, scalable, and easy to troubleshoot.

VLAN 10	CHECKIN	10.10.10.0/24	Gateway 10.10.10.1
VLAN 20	GATE	10.10.20.0/24	Gateway 10.10.20.1
VLAN 30	BAGGAGE	10.10.30.0/24	Gateway 10.10.30.1
VLAN 40	AIRPORT_OPS	10.10.40.0/24	Gateway 10.10.40.1
VLAN 50	IT_MGMT	10.10.50.0/24	Gateway 10.10.50.1
VLAN 60	GUEST_WIFI	10.10.60.0/24	Gateway 10.10.60.1
VLAN 70	SERVERS	10.10.70.0/24	Gateway 10.10.70.1

Step 3: I configured VLANs on the core switch

I created VLANs for each department and assigned names that match the business function.

```
vlan 10
name CHECKIN
vlan 20
name GATE
vlan 30
name BAGGAGE
vlan 40
name AIRPORT_OPS
vlan 50
name IT_MGMT
vlan 60
name GUEST_WIFI
vlan 70
name SERVERS
```

Step 4: I assigned switch ports as access ports

Endpoint-facing ports were configured as access ports. This means the endpoint belongs to one VLAN only.

```
interface Ethernet0/1
description CHECKIN_PC1_VLAN10
switchport mode access
switchport access vlan 10
spanning-tree portfast

interface Ethernet0/3
description BAG_SCANNER1_VLAN30
switchport mode access
switchport access vlan 30
spanning-tree portfast
```

Step 5: I configured the switch-to-router trunk

The link between the switch and router must carry multiple VLANs. I used 802.1Q trunking so one physical cable could carry all VLAN traffic.

```
interface Ethernet0/0
description TRUNK_TO_R1_AIRPORT_EDGE
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40,50,60,70
```

Step 6: I configured router-on-a-stick

Because the switch is Layer 2, it cannot route between VLANs. I used router-on-a-stick so one router interface could support multiple subinterfaces. Each subinterface acts as the default gateway for one VLAN.

```
interface FastEthernet0/0
no shutdown

interface FastEthernet0/0.10
description CHECKIN_GATEWAY
encapsulation dot1Q 10
ip address 10.10.10.1 255.255.255.0

interface FastEthernet0/0.30
description BAGGAGE_GATEWAY
encapsulation dot1Q 30
ip address 10.10.30.1 255.255.255.0

interface FastEthernet0/0.70
description SERVER_GATEWAY
encapsulation dot1Q 70
ip address 10.10.70.1 255.255.255.0
```

Step 7: I created DHCP pools per VLAN

I configured the router to automatically give clients the correct IP address, gateway, DNS server, and domain name.

```
service dhcp
ip dhcp excluded-address 10.10.10.1 10.10.10.20
ip dhcp excluded-address 10.10.30.1 10.10.30.80
ip dhcp excluded-address 10.10.70.1 10.10.70.30

ip dhcp pool VLAN10_CHECKIN
network 10.10.10.0 255.255.255.0
default-router 10.10.10.1
dns-server 10.10.70.10
domain-name airport.local

ip dhcp pool VLAN30_BAGGAGE
network 10.10.30.0 255.255.255.0
default-router 10.10.30.1
dns-server 10.10.70.10
domain-name airport.local
```

Step 8: I added an internal DNS design

The DNS server is in VLAN 70 at 10.10.70.10. DHCP gives that DNS server to all departments. That way, users and devices can use names instead of memorizing IP addresses.

Example DNS records:

```
dns.airport.local      -> 10.10.70.10
apps.airport.local     -> 10.10.70.20
baggage.airport.local  -> 10.10.70.30
printer-bag1.airport.local -> 10.10.30.50
```

Step 9: I configured ACLs for security

VLANs separate networks, but once routing is enabled, VLANs can potentially communicate. ACLs enforce the security policy.

Guest Wi-Fi should not reach internal systems:

```
ip access-list extended GUEST-WIFI-IN
5 permit udp any eq 68 any eq 67
10 deny ip 10.10.60.0 0.0.0.255 10.10.0.0 0.0.255.255
20 permit ip 10.10.60.0 0.0.0.255 any
```

Baggage devices should reach only required services:

```
ip access-list extended BAGGAGE-IN
5 permit udp any eq 68 any eq 67
10 permit udp 10.10.30.0 0.0.0.255 host 10.10.70.10 eq domain
20 permit tcp 10.10.30.0 0.0.0.255 host 10.10.70.30 eq 443
30 permit icmp 10.10.30.0 0.0.0.255 10.10.70.0 0.0.0.255
40 deny ip 10.10.30.0 0.0.0.255 10.10.60.0 0.0.0.255
50 permit ip 10.10.30.0 0.0.0.255 any
```

Then I applied ACLs inbound on the router subinterfaces:

```
interface FastEthernet0/0.30
ip access-group BAGGAGE-IN in

interface FastEthernet0/0.60
ip access-group GUEST-WIFI-IN in
```

Step 10: I hardened management access with SSH

I used SSH because Telnet sends credentials in clear text. In a real airport or enterprise environment, management traffic should be encrypted and restricted.

```
hostname R1-AIRPORT-EDGE
ip domain-name airport.local
username netadmin privilege 15 secret StrongPassword123!
crypto key generate rsa modulus 2048
ip ssh version 2
service password-encryption
enable secret EnableStrongSecret123!

line vty 0 4
login local
transport input ssh
exec-timeout 10 0
```

Optional management restriction:

```
ip access-list standard MGMT-SSH-ONLY
permit 10.10.50.0 0.0.0.255
deny any

line vty 0 4
access-class MGMT-SSH-ONLY in
```

5. Why I Used Each Networking Concept

VLANs

I used VLANs to separate departments into different broadcast domains and security zones. This reduces broadcast traffic, improves security, and makes troubleshooting easier. If baggage devices have a problem, I can focus on VLAN 30 instead of checking the whole network.

802.1Q Trunking

I used trunking because many VLANs needed to cross one link between the switch and router. Without trunking, I would need separate physical router interfaces for every VLAN, which is not scalable.

Router-on-a-Stick

I used router-on-a-stick because a Layer 2 switch cannot route between VLANs. The router subinterfaces provide gateways for each VLAN and allow controlled communication between departments.

DHCP

I used DHCP because manually configuring every airport endpoint would be slow and error-prone. DHCP automatically gives each endpoint the correct IP address, subnet mask, gateway, DNS server, and domain.

Internal DNS

I used DNS because users and devices should use service names, not memorize IP addresses. If a

server IP changes, the DNS record can be updated centrally.

ACLs

I used ACLs to enforce security policy. VLANs create separation, but ACLs decide what traffic is allowed or denied after routing is enabled. This is especially important for guest Wi-Fi and baggage systems.

SSH Management

I used SSH because network device management should be encrypted. I also separated IT management into its own VLAN to show the principle of secure administration.

6. Main Challenge I Faced: DHCP Failed Because of ACLs

The biggest challenge in the lab was that some VPCS clients did not receive DHCP addresses.

At first, the network looked correct:

- VLANs existed on the switch.
- Access ports were assigned to the correct VLANs.
- The trunk was carrying VLANs 10,20,30,40,50,60,70.
- Router subinterfaces were up/up.
- DHCP pools were configured.

But some clients still did not get IP addresses.

Troubleshooting process

I followed a layered troubleshooting method:

```
show ip interface brief
show vlan brief
show interfaces trunk
show ip dhcp pool
show ip dhcp binding
show access-lists
show running-config | include access-group
```

Root cause

Inbound ACLs were applied to some router subinterfaces. The ACLs allowed traffic from normal subnet addresses like 10.10.30.0/24, but DHCP clients do not have an IP address yet when they send their first DHCP Discover message.

DHCP Discover uses:

```
Source IP: 0.0.0.0
Destination IP: 255.255.255.255
UDP source port: 68
UDP destination port: 67
```

So the ACL did not match the pre-DHCP traffic and blocked the request.

Fix

I added this permit line near the top of each affected inbound ACL:

```
permit udp any eq 68 any eq 67
```

Example:

```
ip access-list extended CHECKIN-IN  
5 permit udp any eq 68 any eq 67
```

After applying the fix, I cleared the VPCS IP and requested DHCP again:

```
clear ip  
ip dhcp  
show ip
```

Then the clients received correct IP addresses, gateways, DNS server, and domain name.

What this challenge proves

This challenge proves that I do not just copy commands. I understand packet flow and troubleshooting. I checked Layer 1/2 first, then VLANs, trunking, Layer 3 gateways, DHCP, and finally ACL behavior. That is the same method a NOC or network support team would use during a real incident.

7. Baggage Scanner and Printer Troubleshooting Story

I also included baggage devices because airport operations depend on scanners and printers. If a scanner cannot reach the baggage application server, or a printer cannot print tags, flights and passengers can be delayed.

For a baggage scanner issue, I would troubleshoot like this:

1. Check the endpoint IP information.
2. Ping the default gateway 10.10.30.1.
3. Ping the DNS server 10.10.70.10.
4. Test reachability to the baggage app server 10.10.70.30.
5. Verify the switch port is in VLAN 30.
6. Check trunk forwarding.
7. Check BAGGAGE-IN ACL counters.
8. Confirm the printer IP, subnet mask, gateway, and VLAN.

Commands:

```
show interfaces status
show running-config interface e0/3
show vlan brief
show access-lists BAGGAGE-IN
```

This part of the project shows practical support thinking. I connected technical commands to a real business problem.

8. End-to-End Verification Plan

After building the network, I verified each layer.

Layer 1 and Layer 2

```
show interfaces status
show vlan brief
show interfaces trunk
```

I confirmed that access ports were connected, VLANs were created, and the trunk carried all required VLANs.

Layer 3 routing

```
show ip interface brief
show ip route
```

I confirmed that router subinterfaces were up/up and that each VLAN subnet appeared as a connected route.

DHCP

```
show ip dhcp pool
show ip dhcp binding
```

On each VPCS endpoint:

```
clear ip
ip dhcp
show ip
ping 10.10.x.1
ping 10.10.70.10
```

Security testing

I tested the security policy:

- Guest Wi-Fi to internal server should be blocked.

- Baggage to DNS should be allowed.
- Baggage to baggage app should be allowed.
- Check-in to DNS should be allowed.
- IT management to router SSH should be allowed.
- Non-IT VLAN to router SSH should be denied if the management ACL is configured.

9. Full HR-Friendly Interview Answer

If HR asked me to explain the project step by step, I would say:

> This project was an EVE-NG airport network simulation. I wanted to show that I understand how enterprise networks are designed, not just how to ping between devices. I created a realistic airport environment with separate departments: check-in, gate, baggage, airport operations, IT management, guest Wi-Fi, and servers.

>

> I started by creating a VLAN and IP plan. Each department received its own VLAN and subnet. I did this because different departments have different security and operational needs. For example, guest Wi-Fi should not be in the same network as baggage systems or internal servers.

>

> On the switch, I created the VLANs and assigned endpoint ports as access ports. Then I configured the switch-to-router link as an 802.1Q trunk. I used trunking because one physical link needed to carry multiple VLANs to the router.

>

> On the router, I used router-on-a-stick. That means one physical router interface is divided into subinterfaces, and each subinterface is tagged for a specific VLAN. Each subinterface acts as the default gateway for that VLAN.

>

> I configured DHCP pools for each department so endpoints automatically received the correct IP address, gateway, DNS server, and domain name. I placed the DNS server in the servers VLAN and gave its address to clients through DHCP.

>

> For security, I used ACLs. Guest Wi-Fi was blocked from internal airport networks, and baggage traffic was limited to required services like DNS and the baggage application server. I also configured SSH management so devices could be managed securely instead of using Telnet.

>

> The main challenge I faced was DHCP failing on some VLANs. I troubleshot it step by step. I checked router interfaces, switch VLANs, trunking, DHCP pools, DHCP bindings, and ACLs. I found that inbound ACLs were blocking DHCP Discover traffic because DHCP clients use 0.0.0.0 before they receive an IP address. I fixed it by permitting UDP source port 68 to destination port 67 near the top of the ACLs. After that, the clients received IP addresses successfully.

>

> What I learned from this project is how to think like a network support technician: understand the business need, design segmentation, apply security controls, verify each layer, and troubleshoot with evidence instead of guessing.

10. Short 60-Second Interview Pitch

> I built an EVE-NG airport VLAN network to simulate a real enterprise environment. I separated airport departments into VLANs for check-in, gate, baggage, operations, IT management, guest Wi-Fi, and servers. I configured switch access ports, 802.1Q trunking, router-on-a-stick gateways, DHCP scopes, DNS, ACL security policies, and SSH management. The biggest issue I solved was DHCP failure caused by inbound ACLs blocking DHCP Discover traffic. I identified the problem by checking VLANs, trunks, router subinterfaces, DHCP pools, bindings, and ACLs. Then I fixed it by permitting UDP 68 to UDP 67 before restrictive ACL rules. This project demonstrates VLAN design, routing, DHCP, security, and troubleshooting skills for NOC, IT support, and junior network roles.

11. Technical Interview Questions and Strong Answers

Question: Why did you use VLANs?

I used VLANs to separate different airport departments into different broadcast domains and security zones. This improves security, reduces unnecessary broadcast traffic, and makes troubleshooting easier.

Question: What is 802.1Q trunking?

802.1Q trunking allows multiple VLANs to travel over one physical link by adding VLAN tags to Ethernet frames. In my lab, the trunk between the switch and router carries VLANs 10 through 70.

Question: What is router-on-a-stick?

Router-on-a-stick is a design where one router interface uses multiple subinterfaces. Each subinterface has an 802.1Q VLAN tag and an IP address that acts as the default gateway for that VLAN.

Question: Why did DHCP fail?

DHCP failed because inbound ACLs were blocking DHCP Discover traffic. Before a client receives an IP address, it sends traffic from 0.0.0.0 using UDP source port 68 to destination port 67. The fix was to permit DHCP traffic before the restrictive ACL rules.

Question: How did you secure guest Wi-Fi?

I placed guest Wi-Fi in its own VLAN and applied an ACL that denies access to internal 10.10.0.0/16 networks while allowing permitted outbound traffic. That keeps guest users away from airport systems.

Question: How would you troubleshoot a baggage printer that cannot print?

I would check the endpoint IP, gateway, DNS, switch port VLAN, trunk, and ACL counters. I would ping the gateway first, then DNS, then the baggage application server or printer. If ACL counters

increase, I would review whether the required traffic is permitted.

12. Resume and LinkedIn Bullet Points

Use honest lab/project wording:

- Built an EVE-NG airport VLAN network lab using VLAN segmentation, 802.1Q trunking, router-on-a-stick inter-VLAN routing, DHCP scopes, internal DNS, ACL security policies, and SSH management hardening.
- Configured department-based VLANs for check-in, gate operations, baggage, airport operations, IT management, guest Wi-Fi, and servers using a structured IP addressing plan.
- Troubleshooted DHCP failures by verifying switch access ports, trunk forwarding, router subinterfaces, DHCP pools, DHCP bindings, and inbound ACL behavior.
- Resolved DHCP failure caused by ACLs blocking DHCP Discover traffic by permitting UDP client port 68 to server port 67 before restrictive ACL rules.
- Documented baggage scanner/printer troubleshooting workflow using gateway tests, DNS reachability, ACL counters, and switch port verification.

13. Job-Ready Closing Statement

This project is valuable because it combines technical configuration with business reasoning. I did not only create VLANs; I explained why each VLAN exists, how traffic moves through the switch and router, how IP addressing is automated, how DNS supports operations, how ACLs protect sensitive systems, and how to troubleshoot when something breaks.

For an IT company, this project shows readiness for NOC Analyst, IT Support Technician, Network Support Technician, Junior Network Administrator, and Cloud/Network Support roles because it demonstrates hands-on configuration, structured troubleshooting, documentation, and security-minded thinking.